Phần B: Bitcoin, Hợp đồng thông minh, Tài chính phi tập trung, Web3

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Tiền bitcoin sinh ra từ đâu?



https://www.blockchain.com/explorer/blocks/btc/0 hoặc https://bitaps.com/0

- Bitcoin is against infinitely money-print by central banks and inflation
- Bitcoin is decentralized P2P network for store of value and money transfer

Bitcoin block 0 mined by Anonymus

Time	2009-01-03 18:15:05
Transactions	1
Size	285 bytes
Stripped size	285 bytes
Weight	1 140
Block difficulty	2 536
Network difficulty	1
Version	0x01
Bits	0x1d00ffff
Nonce	0x7c2bac1d
Block reward	50.00000000 BTC
Fee reward	0 BTC

Coinbase message

EThe Times 03/Jan/2009 Chancellor on brink of second bailout for banks

Block	000000000019d6689c085ae165831e934ff763ae46a2a6c172b3f1b60a8ce26f
Previous block	0000000000000000000000000000000000000000000000000000000000000000
Next block	00000000839a8e6886ab5951d76f411475428afc90947ee320161bbf18eb6048
Merkle root	4a5e1e4baab89f3a32518a88c31bc87f618f76673e2cc77ab2127b7afdeda33b

Coinbase hex

```
04ffff001d0104455468652054696d65732030332f4a616e2f3230303920436861e63656c6c6f72206f6e206272696e6b206f662073
65636f6e64206261696c6f757420666f722062616e6b73
```

Header

```
010000000000000000000000000000000000000000000000000000000000000000000000
003ba3edfd7a7b12b27ac72c3e67768f617fc81bc3888a51323a9fb8aa4b1e5e4a29ab
5f49ffff001d1dac2b7c01
```

Tiền bitcoin sinh ra từ đâu?



https://www.blockchain.com/explorer/blocks/btc/0 hoặc https://bitaps.com/0

- Coinbase = Block reward: new generated Bitcoin
- Bitcoin halving: reward per block 50 / 25 / 12.5 / 6.25 / 3.125
- Bitcoin limit: total max supply 21,000,000 BTC
- Current supply: 19M BTC

896922 Binance.com

000000000000000000f6155a5460b0c33d3e0259cfbf1f4c0abe7160adac32

binance/.!HSmmk9_czlo+#J4rf1K TV

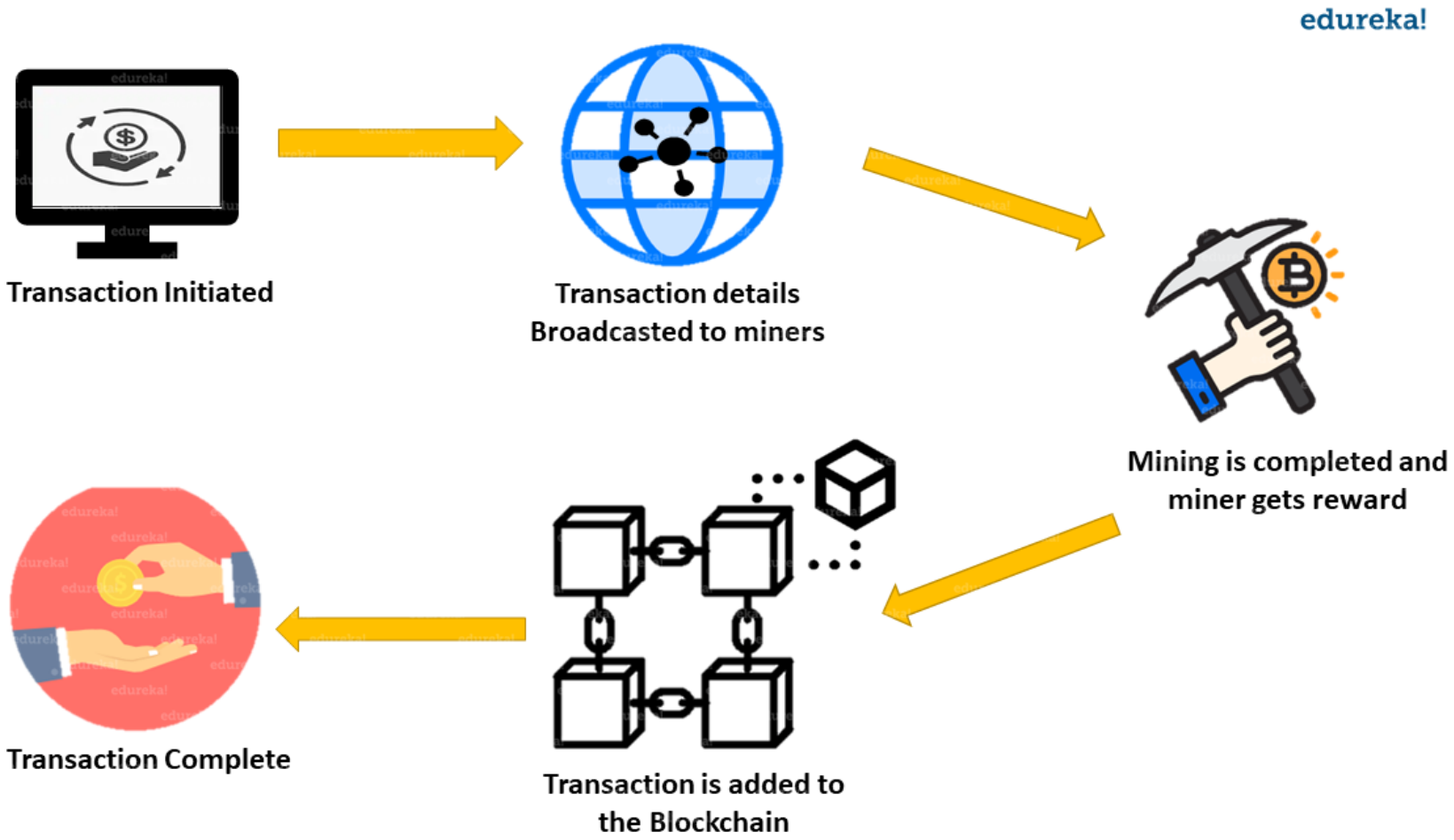
Average fee rate: 3.21 ś/vByte

Transactions	1 773	Version	0x20066000
Size	1 639 677 bytes	Bits	0x17025ced
Stripped size	784 729 bytes	Nonce	0x8bc4001f
Weight	3 993 864	Block reward	3.12500000 BTC
Difficulty	292 813 297 134 157	Fee reward	0.03206580 BTC

Cơ chế đào bitcoin?

https://www.blockchain.com/explorer/charts/hash-rate

- Block reward
- Fee reward
- Mining



Cơ chế đào bitcoin?

https://www.blockchain.com/explorer/charts/hash-rate

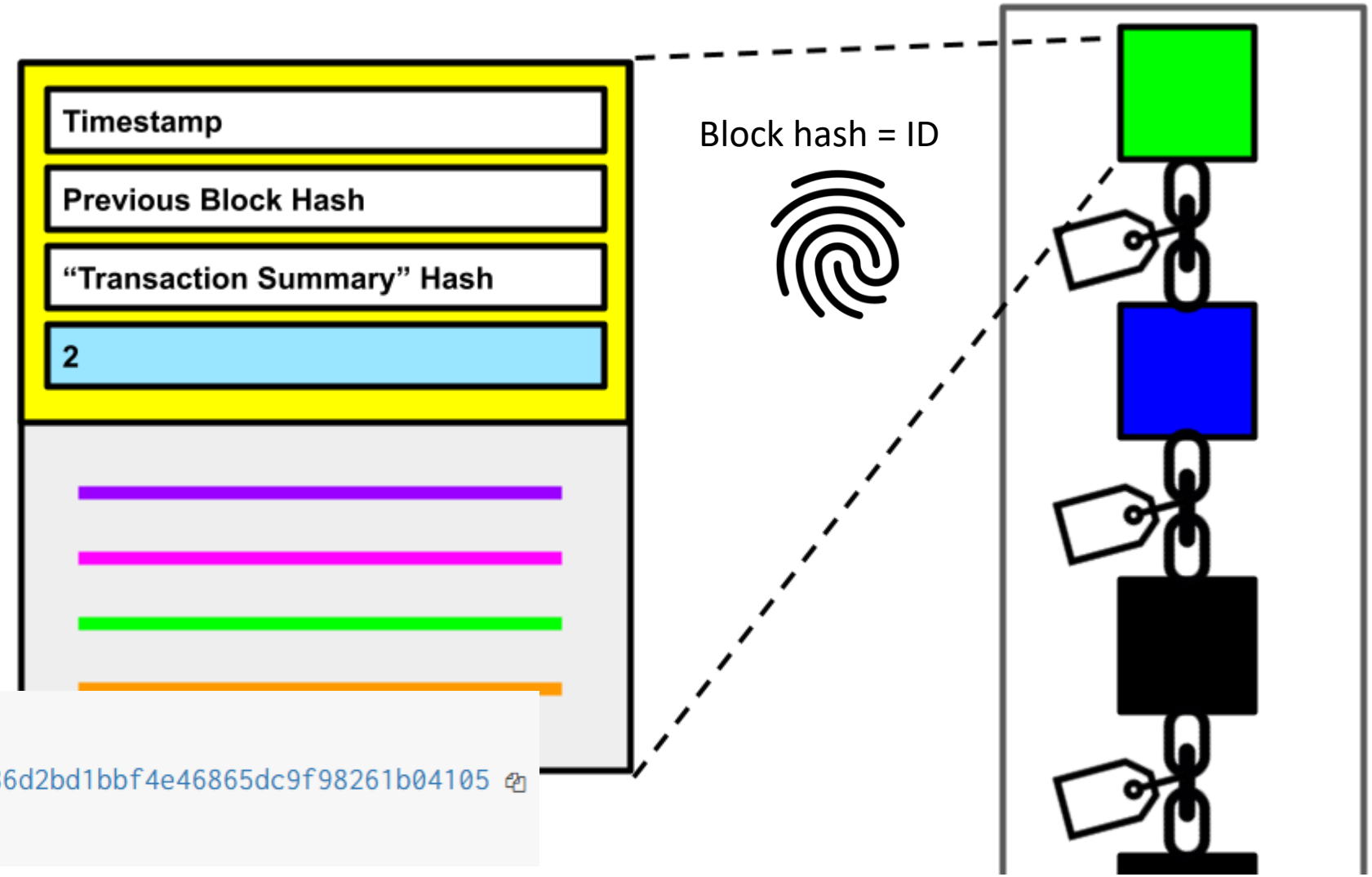
Block hash = ***HASH*** (timestamp, prev.blockhash, txhash, nonce)

- SHA-256
- Nonce
- Mining
- Target difficulty

Target 3 zeros: 000

➢ Valid: 000… , 0000…

➢ Invalid: 00…



896938 AntPool

0000000000000000001352419f13f8486d2bd1bbf4e46865dc9f98261b04105

Mined by AntPool So.mm)-[Tc`Y>)geX0{_}8]G0]?

Đọc giao dịch Bitcoin

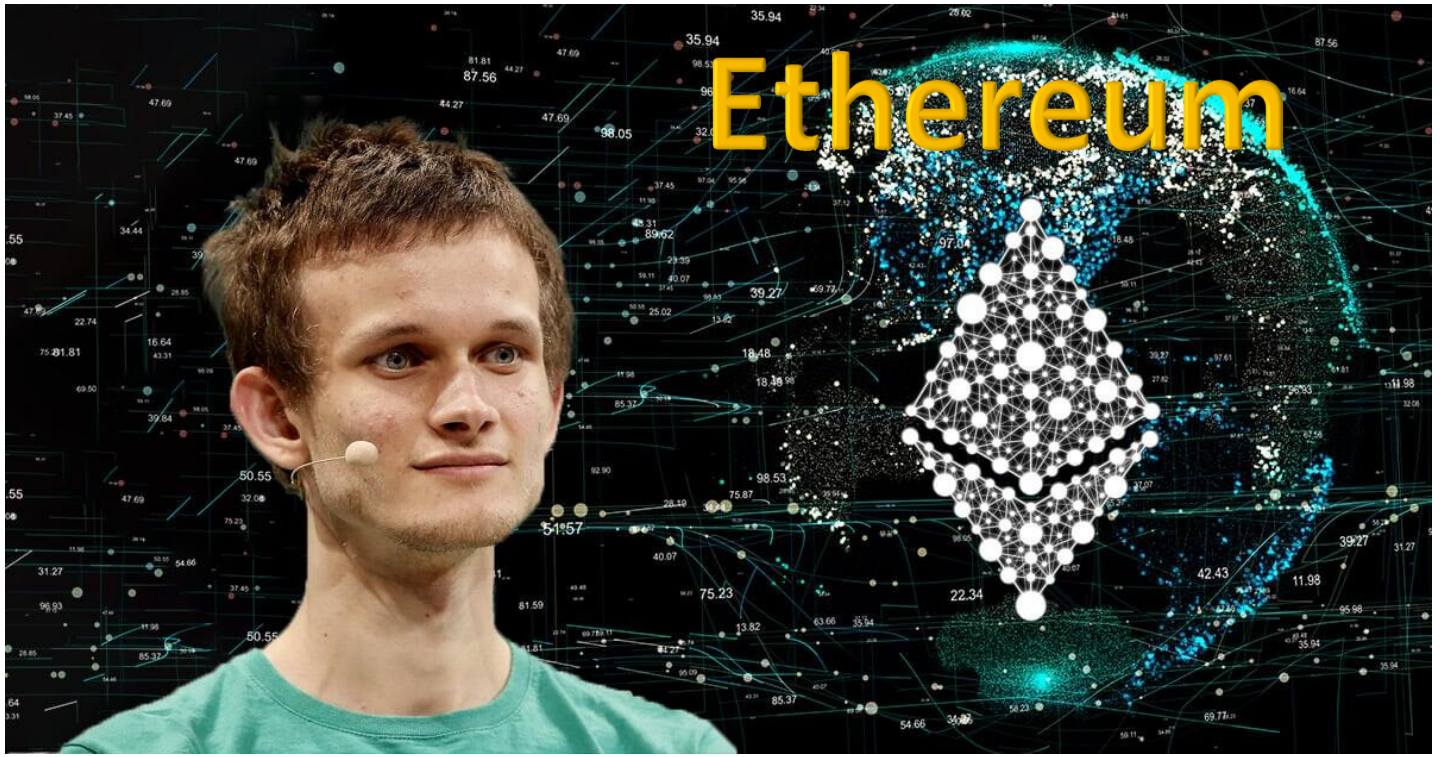
Ethereum: blockchain thế hệ 2

- From Bitcoin scripts to a Turing-complete & programmable blockchain
- Smartcontract, decentralized finance/identity/application
- Smartcontract platform
- World computer

✓ Vitalik Buterin (1994)
✓ Gavin Wood (1980)
✓ Ethereum Virtual Machine (EVM)
✓ PoW to PoS

https://ethereum.org/vi/whitepaper/

https://ethereum.github.io/yellowpaper/paper.pdf



Smartcontract & dapps

- A contract = rules to be executed
- EVM-contract
- Solidity language
- A decentralized application (dapp) = multi-contracts deployed on EVM



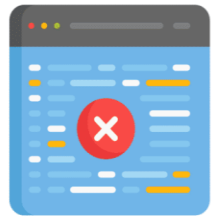


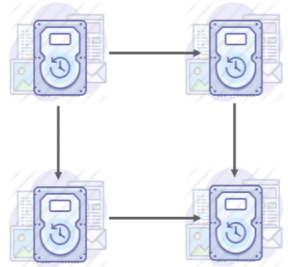












dapps | contracts

EVM runtime

Ethereum Network

ERC-standards

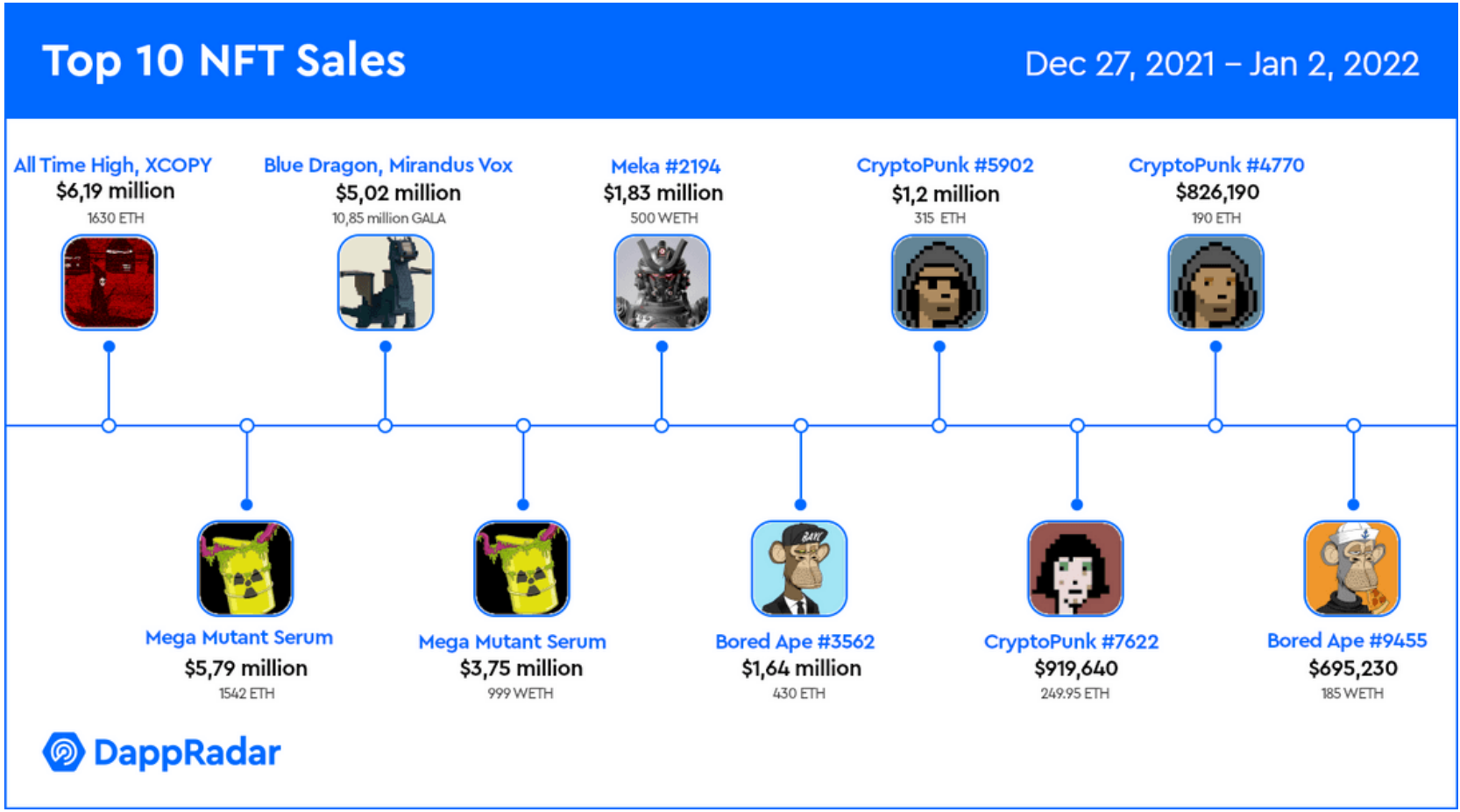
ERC = Ethereum Request for Comment

ERC-20: fungible tokens for accounting

ERC-721: non-fungible token (NFT)

Examples:

- ERC-20: USDT, USDC (**$126B**)
- ERC-721: NFT artworks



Cài đặt ví Metamask và giao dịch ETH testnet

- Install Metamask extension
- Search Sepolia testnet
- Add Sepolia testnet to Metamask

Phát hành token (tiền) mã hóa

Thực hành

- Lập trình smartcontract
- Triển khai contract ERC20 phát hành token
- Triển khai contract ERC721 phát hành NFT